JP's Portfolio

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Round 2: Portfolio Construction Challenge

1 Problem and Strategy Overview

Round 2 asks each team to construct and manage a maximum 10-stock equity portfolio drawn from the Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100 universes, backtested from 1 January 2021 to 31 December 2025 with a virtual corpus of Rs 1 crore. The primary ranking metric is Total Net PnL, and a forward-looking out-of-sample stress test on 1 January to 30 June 2026 checks that the strategy generalises beyond the backtest window.

The central idea is a single, fully systematic composite-factor strategy applied identically to every eligible stock: a quarterly-rebalanced ranking on three price-based factors (12-1 month momentum, low-volatility, and a trend-quality measure), sized by inverse volatility with a concentration cap. A daily 200-day moving-average trend filter overlays this quarterly cycle for additional risk management. The design was deliberately kept simple and fully rules-based, so another evaluator can trace exactly how the portfolio was produced at any point in time. It was also validated using a train/test split and a rolling-window robustness check, rather than chosen purely for the highest in-sample return, in anticipation of the forward stress test.

2 Data

Universe. The eligible universe (≈300 stocks) was built by combining the current NSE constituent lists for Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100. A handful of symbols appear on more than one list; these are de-duplicated to a single entry recording every index the stock belongs to. Since point-in-time historical index membership is not freely available, the current constituent lists are applied retroactively across the full 2021–2025 backtest; this is a stated simplification (see Limitations).

Prices. Daily OHLCV data for every universe stock and for the two benchmark indices was downloaded via the `yfinance` package, covering June 2019 onward (well before the 2021 backtest start) through the end of the backtest. Prices are auto-adjusted closes (dividend- and split-adjusted). Only the adjusted *close* column feeds the model itself; open/high/low/volume are downloaded but not used by the current strategy. The extra pre-2021 history is a buffer so the 252-day momentum/quality lookback and the 200-day trend filter have sufficient data from day one of the backtest. At each quarterly rebalance, any stock with less than 95% price coverage over the lookback window (mostly recent IPOs) is excluded from that quarter's ranking rather than treated as having a zero or missing score.

Fundamental data. No fundamental data (earnings, book value, ROE, etc.) is used. This is a deliberate choice, not an oversight: reliable, point-in-time fundamental data free of look-ahead bias is not readily available for 300 stocks spanning large, mid and small caps in India. Building the model on price data alone also keeps the same formula applicable to every stock without exception.

3 Methodology

3.1 Stock selection

Every quarter (first trading day of Jan/Apr/Jul/Oct), a composite score is computed for every eligible stock:

$$\text{Composite} = 0.5 \times \left[0.5\, Z(\text{Momentum}) + 0.5\, Z(\text{Low-Vol})\right] + 0.5\, Z(\text{Quality})$$

Here, **Momentum** is the 12-month price return skipping the most recent month (the standard "12-1" construction, which avoids short-term reversal effects). **Low-Vol** is the negative of the trailing 90-day daily return standard deviation. **Quality** is the R^2 of a linear trend fit to log-price over the same 12-month window, which penalises spiky, parabolic price paths in favour of smoothly-trending ones. Each factor is cross-sectionally z-scored across all stocks with at least 95% price history over the lookback window; stocks without enough history (mostly recent IPOs) are automatically excluded that quarter. The 10 highest composite-score stocks are then selected.

Why these three factors. Momentum and low-volatility are two of the most extensively documented factors in equity research, and they are deliberately complementary rather than redundant. Momentum drives the return by following stocks that are already trending, while low-volatility manages risk, since momentum alone is known to be prone to sharp reversals (a "momentum crash") after a strong run. The quality/trend-smoothness factor was added for a specific, tested reason rather than assumed upfront. An earlier two-factor version of this strategy underperformed the benchmark in a blind 2025 test (Section 7), and the cause was concentration in momentum names with erratic rather than steady price paths–exactly the names most prone to reversal. Adding trend-quality was confirmed, not merely assumed, to fix that weakness (Section 7). Value and fundamental-quality factors were excluded for the data-reliability reason above. Size is already handled structurally by the universe spanning three cap segments. Liquidity was not added separately since it overlaps heavily with volatility. And a short-term reversal factor was rejected because it would work against momentum over a similar horizon.

3.2 Weighting

Selected stocks are sized inverse to their own trailing 90-day volatility (the same volatility used in the Low-Vol factor), so that size is set by each stock's own risk rather than by how high it ranks on the noisier composite score. Raw inverse-volatility affinities are shifted down by their minimum (plus a small constant) before normalising, then capped at 15% of portfolio value per name. Any excess above the cap is redistributed proportionally among the remaining names, iterated until no name exceeds the cap. This keeps a single very low-volatility name from dominating the book.

3.3 Rebalancing and risk management

Stock selection and target weights are decided quarterly. Only the delta needed to reach the new target is traded for names that remain selected, while dropped names are fully sold and new entrants are bought to target. Independent of this quarterly schedule, a 200-day moving-average trend filter is checked *daily* on every held position. If a stock's close falls below its own 200-day average, the position is fully exited to cash (a defensive rule, not a directional bet). If price reclaims the average before the next rebalance, the position is re-bought using the cash reserved for that slot. A flat 0.1% transaction cost is charged on the notional value of every executed buy and sell, and is deducted directly from cash. This means it compounds into every subsequent decision rather than being a cosmetic adjustment applied at the end.

3.4 Consistency and parameter selection

The same formula, thresholds and rules are applied to every one of the ≈300 eligible stocks with no per-stock discretion. The one free parameter with any real tuning behind it–the 50% weight given to the quality factor–was selected using only 2021–2024 training data and full-backtest metrics, deliberately excluding the 2025 held-out test, via a 0.001-resolution grid search. That search confirmed the chosen weight sits inside a genuine stable plateau (0.499–0.503, five consecutive 0.001-steps giving bit-identical results) rather than an isolated, fragile spike. This distinction matters because it is easy to overfit a single free parameter to one backtest window.

3.5 Key assumptions

- Trades execute at the same-day closing price as the signal date (standard close-to-close convention).
- All trades are in whole shares (no fractional shares), with target rupee allocations floored to the nearest full share.
- Idle or exited cash earns 0%.
- Corporate actions are handled via Yahoo Finance's auto-adjusted close series.
- No slippage, shorting or leverage is modelled beyond the stated 0.1% transaction cost.

4 Tools and Software Used

The entire pipeline is written in Python 3, using `pandas` and `numpy` for data handling and the day-by-day backtest simulation, and `yfinance` for price data acquisition. No machine-learning or numerical-optimisation library is used: the model is a transparent, rules-based factor score rather than a fitted or black-box one. This was a deliberate choice, given the requirement that another evaluator be able to understand exactly how the portfolio was produced. `matplotlib` was used for the chart in this report, and the report itself is typeset in LaTeX. The complete code is in the accompanying GitHub repository (see the README for how to reproduce every number in this report).

5 Results and Performance Metrics

Metric	**Value**
Final Portfolio Value	Rs 6.76 crore
Total Net PnL	Rs 5.76 crore
Absolute Return	576.23%
Annualised Return (CAGR)	46.60%
Maximum Drawdown	−23.78%
Sharpe Ratio (rf = 0%)	2.66
Gain-to-Loss Ratio	2.80
Accuracy (% profitable closed trades)	52.59%
Total Executed Orders	347 (179 buys, 168 sells)
Closed Round-Trip Trades	135
Average Trades per Stock	3.94
Turnover (cumulative, 5-year)	94.3× initial capital
Total Transaction Costs Paid	Rs 9.43 lakh

Table 1: Required Round 2 metrics, full 2021–2025 backtest.

For additional context beyond the required list, the strategy also shows Beta ≈ 0.85 and annualised Alpha ≈ +33.6% against Nifty 500, an Information Ratio of 2.45, a Sortino Ratio of 3.86, and a Calmar

Ticker	Company	Shares	Price (Rs)	Weight
BHARTIARTL	Bharti Airtel Ltd.	5,028	2,079.43	15.46%
NAVINFLUOR	Navin Fluorine International Ltd.	1,636	5,912.89	14.31%
EICHERMOT	Eicher Motors Ltd.	1,334	7,236.72	14.28%
FORTIS	Fortis Healthcare Ltd.	9,382	883.06	12.25%
LAURUSLABS	Laurus Labs Ltd.	6,774	1,106.90	11.09%
MUTHOOTFIN	Muthoot Finance Ltd.	1,867	3,779.74	10.44%
MANAPPURAM	Manappuram Finance Ltd.	18,765	306.66	8.51%
ASTERDM	Aster DM Quality Care Ltd.	8,011	614.16	7.28%
FORCEMOT	Force Motors Ltd.	1	20,566.00	0.03%
Cash	(uninvested, awaiting redeployment)	–	–	6.37%

Table 2: Final portfolio composition and weights, as of 31 December 2025. Only 9 of the permitted 10 slots are filled: the 10th position went unfilled that quarter when sequential rebalance trades exhausted available cash before its turn (Section 7), leaving that slice as cash rather than a modelling error.

Ratio of 1.96. Beta below 1 shows the outperformance is not simply leveraged market exposure, and $R^2 \approx 0.49$ shows that a large share of the return comes from stock selection rather than broad market movement.

6 Benchmark Comparison

Benchmark choice. Nifty 500 (index `^CRSLDX`) is used as the primary benchmark because the eligible universe spans large, mid and small caps. A broad, ≈500-name, all-cap index is a more representative comparator than a large-cap-only index such as Nifty 100. Nifty 100 is reported alongside for completeness, since the challenge permits either.

	Strategy	Nifty 500	Nifty 100
CAGR	46.60%	15.36%	13.35%
Total Return, 2021–2025	576.23%	104.17%	87.07%
Maximum Drawdown	−23.78%	−18.84%	−17.56%
Sharpe Ratio (rf = 0%)	2.66	1.07	0.94
Outperformance (CAGR, pp)	–	+31.24	+33.25

Table 3: Strategy vs. benchmark, full 2021–2025 window.

The strategy outperforms both benchmarks by a wide margin on CAGR and total return, at the cost of a somewhat deeper maximum drawdown than Nifty 500–expected for a concentrated 10-stock book relative to a ≈500-name index, and discussed further below.

7 Limitations and Discussion

Concentration and momentum-crash risk. A 10-stock, trailing-momentum-tilted portfolio is structurally more exposed to leadership-rotation risk than a broad index. This is not theoretical: when the quality factor was withheld and the strategy re-run as a blind, walk-forward test (rules locked on 2021–2024, evaluated unmodified on 2025 alone), the two-factor version returned −6.4% to −11.1% in 2025, while Nifty 500 returned +4.6% and Nifty 100 +6.6%. Only the final three-factor design was profitable and beat both benchmarks in that same blind test (+7.9%). This is encouraging evidence the fix works, but it is a single held-out year and does not guarantee the strategy is immune

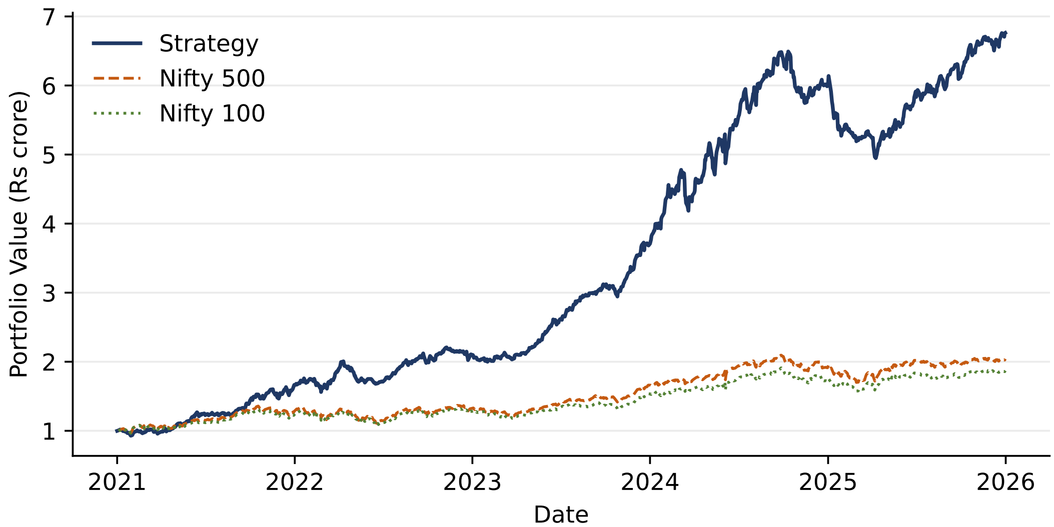

Figure 1: Portfolio value vs. Nifty 500 and Nifty 100, 2021–2025 (all series start at Rs 1 crore).

to a different kind of reversal in the future.

A tested and rejected risk control. A portfolio-level drawdown circuit breaker (de-risking 50% of the book whenever it fell more than 8% from its post-rebalance peak) was tested against a 20-quarter rolling grid. It shrank the worst single quarter from −16.2% to −14.5%, but it also reduced the quarterly hit-rate against the benchmark from 90% to 85% and cut every other headline metric in the official 2021–2025 backtest at once (Net PnL from Rs 5.76cr to Rs 4.19cr, CAGR from 46.6% to 39.0%, Sharpe from 2.66 to 2.46, Gain-to-Loss from 2.80 to 2.63), without even improving full-period maximum drawdown. Since Total Net PnL is the primary Round 2 ranking metric, this was a deliberate, metrics-driven decision to exclude the breaker, not an oversight. It reflects a real trade-off between smoother quarter-to-quarter behaviour and the metric the round is actually scored on.

Execution friction. Rebalance trades execute sequentially against a fixed cash pool each quarter. Because target rupee amounts are floored to whole shares, a lower-priority target position can go temporarily unfunded if earlier trades in the same rebalance exhaust available cash–this was observed at least once in the backtest. The unfunded capital sits in cash until the next rebalance rather than being fully deployed. This is a realistic execution constraint rather than a modelling error, but it means realised weights can occasionally differ slightly from target weights.

Universe timing. Using current (rather than historical, point-in-time) index constituent lists for the full 2021–2025 backtest is a simplification driven by data availability; a small number of names in the backtest may not have been constituents of their respective index for the entire period.

No fundamental data. The strategy is intentionally price-only. This avoids a real data-reliability problem for small and mid caps, but it also means the model cannot see valuation, earnings quality, or balance-sheet risk directly. It would not be expected to perform well in a regime where price-momentum and fundamentals diverge sharply.

General caveat. All of the above is based on a single historical path. The out-of-sample stress test (1 January to 30 June 2026) is a genuine test of generalisation rather than a formality. The validation discipline described in Section 3.4–locking parameters before looking at held-out performance–was followed specifically because backtest performance, however strong, is not a guarantee of future results.